

Faculty of Engineering and Architecture

Department of Computer Engineering

CEN 302 – Software Engineering

Police Management System (PMS)

Use Cases Document

Prepared by:

Adorela Dashi

Ester Alisufi

Kaltrina Lala

Amarildo Hazizi

Drini Ziri

Egzon Xheleshi

Petro Boçi

Received by:

M.Sc. Shkumbin Fida

M.Sc. Megi Nako

1. All Users

UC1: Login / Logout

UC1: Login / Logout	
Summary	Any authorized user (staff or citizen) logs into the system using their credentials and logs out when finished.
Dependency	None
Actors	Primary: All Users (Administrator, Police Officer, Detective, Forensic Expert, Traffic Officer, Citizen)
Preconditions	1. User must have valid credentials (staff: issued by Admin; citizens: self-registered). 2. System must be accessible via web browser.
Main Sequence	1. User navigates to the login page. 2. System displays the login form (Username, Password). 3. User enters their username and password. 4. User clicks the Login button. 5. System validates the credentials. 6. System grants access and redirects the user to their role-based dashboard. 7. User performs their tasks. 8. User clicks Logout. 9. System ends the session and returns to the login page.
Alternative Sequence	1. Invalid credentials: System displays an error message. User may retry. 2. Account locked/inactive: System displays an error indicating the account is not active and prompts contacting the Administrator. 3. Citizen account not yet verified: System notifies the citizen to complete registration.
Non-Functional Requirements	1. Login should complete within 3 seconds under normal load. 2. Passwords must be encrypted (hashed) in the database. 3. System must enforce session timeout after inactivity.
Postconditions	1. User is authenticated and their session is active. 2. On logout, the session is terminated and no sensitive data remains accessible.

UC2: View Dashboard

UC2: View Dashboard	
Summary	After logging in, any user can view a role-specific dashboard showing relevant statistics and quick-access actions.
Dependency	UC1 (User logs in)
Actors	Primary: All Users (Administrator, Police Officer, Detective, Forensic Expert, Traffic Officer, Citizen)
Preconditions	1. User is authenticated and logged in. 2. User has a role-specific dashboard configured.
Main Sequence	1. User is redirected to the dashboard after successful login. 2. System retrieves relevant statistics from the database. 3. System displays the dashboard tailored to the role: - Admin: total cases, users per role, departments, incidents - Officer: assigned cases, recent incidents, patrols - Detective: assigned cases, open investigations, suspects

	Forensic Expert: pending analyses, submitted reports - Traffic Officer: accidents, fines, violations - Citizen: submitted reports, fine status, notifications
Alternative Sequence	1. No data available: System displays a message that no statistics are currently available. 2. Data loading error: System shows an error message and provides an option to refresh.
Non-Functional Requirements	1. Dashboard must load within 5 seconds. 2. Statistics must reflect real-time data. 3. Dashboard layout must be responsive and accessible from any device.
Postconditions	1. The dashboard is displayed with up-to-date statistics for the user's role. 2. User can navigate to specific sections from the dashboard.

UC3: Change Own Password

UC3: Change Own Password	
Summary	Any logged-in user can change their own account password for security purposes.
Dependency	UC1 (User logs in)
Actors	Primary: All Users (Administrator, Police Officer, Detective, Forensic Expert, Traffic Officer, Citizen)
Preconditions	1. User is authenticated and logged in. 2. User knows their current password.
Main Sequence	1. User navigates to Account Settings. 2. User selects 'Change Password'. 3. System displays the change password form. 4. User enters current password, new password, and confirms new password. 5. User submits the form. 6. System validates current password and checks new password strength. 7. System updates the password in the database. 8. System confirms with a success message.
Alternative Sequence	1. Incorrect current password: System displays an error and does not update. 2. New password too weak: System prompts the user to choose a stronger password. 3. New password and confirmation do not match: System highlights the mismatch.
Non-Functional Requirements	1. Passwords must meet minimum security requirements (length, complexity). 2. New password must not be the same as the current one. 3. Password change must be logged in the audit trail.
Postconditions	1. Password is updated securely in the system. 2. User session remains active or user is prompted to log in again.

2. Police Officer

UC4: Create Incident Report

UC4: Create Incident Report	
Summary	A Police Officer creates a new incident report to document a reported crime in the system.
Dependency	UC1 (User logs in)
Actors	Primary: Police Officer
Preconditions	1. Police Officer is logged in and has permission to create incident reports. 2. The system is operational.
Main Sequence	1. Officer navigates to the Reports section. 2. Officer clicks 'Create New Incident Report'. 3. System displays the incident report form. 4. Officer enters the required details: location, crime type, description, date and time. 5. Officer submits the form. 6. System validates all required fields. 7. System saves the incident report in the database. 8. System automatically generates a new case linked to the incident report. 9. System displays a confirmation message with the Case ID.
Alternative Sequence	1. Missing required fields: System highlights empty fields and displays an error message. Officer must complete them before submitting. 2. System error during save: System displays an error and prompts the officer to try again.
Non-Functional Requirements	1. The form must be simple and completable within 5 minutes. 2. System must validate data formats (e.g., date/time format, text length). 3. Data must be stored securely with access restricted by role.
Postconditions	1. A new incident report is saved in the system. 2. A new case is automatically created and linked to the report. 3. The case appears in the Cases list with 'Open' status.

UC5: Update Case Status

UC5: Update Case Status	
Summary	A Police Officer updates the status of an existing case (Open, Under Investigation, or Closed).
Dependency	UC1 (User logs in), UC4 (Create Incident Report)
Actors	Primary: Police Officer
Preconditions	1. Officer is logged in. 2. The case exists in the system and is assigned to the officer.
Main Sequence	1. Officer navigates to the Cases section. 2. Officer selects the case to update. 3. System displays the case details. 4. Officer selects a new status from the dropdown (Open, Under Investigation, Closed). 5. Officer saves the update. 6. System validates the status change. 7. System saves the updated status with a timestamp. 8. System confirms the update with a success message.

Alternative Sequence	1. Unauthorized access: If the officer tries to update a case not assigned to them, the system denies the action. 2. Same status selected: System notifies the user that the status is unchanged.
Non-Functional Requirements	1. Status update should be saved within 2 seconds. 2. Status change history should be logged for audit purposes. 3. Access control must prevent unauthorized status changes.
Postconditions	1. The case status is updated in the database. 2. The change is reflected immediately in the case list and dashboard.

UC6: Upload Evidence

UC6: Upload Evidence	
Summary	A Police Officer or Detective uploads evidence files related to an existing case.
Dependency	UC1 (User logs in), UC4 (Create Incident Report)
Actors	Primary: Police Officer Secondary: Detective
Preconditions	1. User is logged in with permission to upload evidence. 2. A case already exists in the system.
Main Sequence	1. User navigates to the specific case. 2. User clicks 'Upload Evidence'. 3. System displays the evidence upload form. 4. User selects the evidence file(s) and enters a description. 5. User submits the upload. 6. System validates the file format and size. 7. System stores the evidence and links it to the case. 8. System confirms the upload with a success message.
Alternative Sequence	1. Invalid file format or size exceeded: System displays an error and requests a valid file. 2. Upload failure: System shows an error and allows retry.
Non-Functional Requirements	1. Supported file formats: PDF, JPG, PNG, MP4, AVI. 2. Maximum file size per upload should be enforced. 3. Evidence files must be stored securely with restricted access.
Postconditions	1. Evidence is stored securely in the system. 2. Evidence is linked to the correct case and visible in case details.

UC7: Record Arrest Information

UC7: Record Arrest Information	
Summary	A Police Officer records the details of an arrest and links them to the relevant case.
Dependency	UC1 (User logs in), UC4 (Create Incident Report)
Actors	Primary: Police Officer
Preconditions	1. Officer is logged in. 2. A case exists in the system. 3. An arrest has been made related to an existing case.

Main Sequence	1. Officer navigates to the relevant case. 2. Officer selects 'Add Arrest Record'. 3. System displays the arrest information form. 4. Officer fills in: arrested person's name, date of arrest, location, charges. 5. Officer submits the form. 6. System validates the input. 7. System saves the arrest record and links it to the case. 8. System confirms with a success message.
Alternative Sequence	1. Incomplete information: System highlights missing required fields. 2. Case not found: System displays an error and prevents submission.
Non-Functional Requirements	1. Arrest records must be immutable once submitted (controlled edit process). 2. Data must comply with law enforcement data regulations. 3. Records must be audit-logged.
Postconditions	1. Arrest record is saved and linked to the case. 2. Case details are updated to reflect the arrest.

UC8: Manage Patrols

UC8: Manage Patrols	
Summary	A Police Officer views, registers, and updates patrol assignments and schedules.
Dependency	UC1 (User logs in)
Actors	Primary: Police Officer
Preconditions	1. Officer is logged in. 2. Patrol data exists or the officer has permission to create new patrols.
Main Sequence	1. Officer navigates to the Patrols section. 2. Officer views the list of current patrols and their status. 3. To register a patrol: Officer clicks 'New Patrol', enters route, time slot, and area. Officer submits the form. System saves the patrol. 4. To update a patrol: Officer selects the patrol, modifies the details, and saves. 5. System confirms each action with a success message.
Alternative Sequence	1. Incomplete patrol data: System highlights missing fields. 2. Patrol time conflict: System warns the officer of a scheduling conflict.
Non-Functional Requirements	1. Patrol data must be saved within 2 seconds. 2. Patrol schedules must be visible to all relevant officers and administrators. 3. All patrol changes must be audit-logged.
Postconditions	1. Patrol record is saved and linked to the officer. 2. Patrol schedule is visible in the officer's dashboard. 3. Changes are reflected immediately for administrators.

UC9: Register Complaint

UC9: Register Complaint	
Summary	A Police Officer registers a complaint received from a citizen or filed in person at the department.

Dependency	UC1 (User logs in)
Actors	Primary: Police Officer
Preconditions	1. Officer is logged in. 2. A complaint has been received or reported in person.
Main Sequence	1. Officer navigates to the Complaints section. 2. Officer clicks 'Register Complaint'. 3. System displays the complaint form. 4. Officer enters: complainant details, complaint description, category, and date. 5. Officer submits the form. 6. System validates the input. 7. System saves the complaint and links it to the relevant person or case (if applicable). 8. System confirms with a success message.
Alternative Sequence	1. Missing required fields: System highlights empty fields. 2. Case not found (when linking to a case): System shows an error.
Non-Functional Requirements	1. Complaint records must be immutable once submitted. 2. Complaint data must be stored securely. 3. Complaints must be audit-logged with timestamp and officer identity.
Postconditions	1. Complaint is saved and accessible in the Complaints section. 2. Complaint is linked to a citizen or case record where applicable.

UC10: Search and Filter Cases

UC10: Search and Filter Cases	
Summary	A Police Officer searches for cases using filters such as status, crime type, date, or case ID.
Dependency	UC1 (User logs in)
Actors	Primary: Police Officer
Preconditions	1. Officer is logged in. 2. One or more cases exist in the system.
Main Sequence	1. Officer navigates to the Cases section. 2. Officer enters search criteria: case ID, crime type, status, or date range. 3. Officer clicks 'Search' or 'Filter'. 4. System queries the database with the given criteria. 5. System displays the filtered list of matching cases. 6. Officer selects a case to view its details.
Alternative Sequence	1. No cases match the criteria: System displays a 'No results found' message. 2. Invalid search input: System highlights the invalid field and requests correction.
Non-Functional Requirements	1. Search results must be returned within 2 seconds. 2. Filters must be combinable (e.g., status AND crime type). 3. Only cases accessible to the officer's role are shown in results.
Postconditions	1. A filtered list of cases matching the search criteria is displayed. 2. Officer can select and open any case from the results.

3. Detective

UC11: Open Criminal Case

UC11: Open Criminal Case	
Summary	A Detective opens a new criminal case or takes ownership of a case linked to an incident report for formal investigation.
Dependency	UC1 (User logs in), UC4 (Create Incident Report)
Actors	Primary: Detective
Preconditions	1. Detective is logged in. 2. An incident report has been submitted by a Police Officer. 3. The case is assigned to the detective.
Main Sequence	1. Detective navigates to the Cases section. 2. Detective selects an existing case or opens a new case linked to an incident report. 3. System displays the case creation / case details form. 4. Detective reviews the incident report and enters additional investigation details. 5. Detective sets the initial status (Open or Under Investigation). 6. Detective submits the case record. 7. System saves the case and links it to the incident report. 8. System confirms with a success message and case ID.
Alternative Sequence	1. Case already open: System notifies the detective that the case already exists and prevents duplication. 2. Linked incident report not found: System shows an error.
Non-Functional Requirements	1. Case creation must complete within 3 seconds. 2. Cases must be linked to an incident report to ensure traceability. 3. All case creation actions must be audit-logged.
Postconditions	1. A criminal case is opened and saved in the system. 2. The case is visible in the detective's case list with the assigned status. 3. The case is linked to the originating incident report.

UC12: Assign Suspect to Case

UC12: Assign Suspect to Case	
Summary	A Detective links a suspect to an existing case by entering suspect details into the system.
Dependency	UC1 (User logs in), UC4 (Create Incident Report)
Actors	Primary: Detective
Preconditions	1. Detective is logged in. 2. A case exists in the system and is assigned to the detective. 3. There is a potential suspect related to the case.
Main Sequence	1. Detective navigates to the relevant case. 2. Detective selects 'Add Suspect'. 3. System displays the suspect information form. 4. Detective enters: name, age, physical description, and associated evidence. 5. Detective submits the form. 6. System validates the input. 7. System saves the suspect information and links it to the case. 8. System confirms with a success message.

Alternative Sequence	1. Case does not exist: System displays an error and prevents submission. 2. Missing required fields: System highlights empty fields and requires completion.
Non-Functional Requirements	1. Suspect data must be stored securely with restricted access. 2. Suspect records must be clearly linked to the correct case. 3. Input validation must prevent invalid data formats.
Postconditions	1. Suspect information is saved and linked to the case. 2. Suspect appears in the case details under the Suspects section.

UC13: Manage Witnesses

UC13: Manage Witnesses	
Summary	A Detective adds, views, and manages witness records and testimonies linked to an existing case.
Dependency	UC1 (User logs in), UC11 (Open Criminal Case)
Actors	Primary: Detective
Preconditions	1. Detective is logged in. 2. A case is assigned to the detective.
Main Sequence	1. Detective navigates to the relevant case. 2. Detective opens the Witnesses section. 3. Detective clicks 'Add Witness'. 4. System displays the witness form. 5. Detective enters: witness name, contact details, testimony summary, and confidentiality status. 6. Detective submits the form. 7. System saves the witness record and links it to the case. 8. To view: Detective selects a witness from the list; system displays full witness profile. 9. System confirms each action with a success message.
Alternative Sequence	1. Missing required fields: System highlights missing fields and prevents submission. 2. Witness already exists for the case: System alerts the detective to avoid duplication.
Non-Functional Requirements	1. Witness data must be stored securely with restricted access. 2. Confidentiality status must determine visibility to other roles. 3. All witness management actions must be audit-logged.
Postconditions	1. Witness record is saved and linked to the case. 2. Witness appears in the case details under the Witnesses section.

UC14: Track Investigation Status

UC14: Track Investigation Status	
Summary	A Detective updates and tracks the investigation status of an assigned case.
Dependency	UC1 (User logs in), UC11 (Open Criminal Case)
Actors	Primary: Detective
Preconditions	1. Detective is logged in. 2. A case is assigned to the detective.

Main Sequence	1. Detective navigates to the Cases section. 2. Detective selects the case to update. 3. System displays current case status and investigation timeline. 4. Detective selects the new status: Open, Under Investigation, or Closed. 5. Detective adds a status update note explaining the change. 6. Detective saves the update. 7. System saves the new status with timestamp and detective identity. 8. System sends a notification to the Administrator. 9. System confirms with a success message.
Alternative Sequence	1. Attempt to close a case without Admin authorization: System prevents the action and displays a message. 2. Same status selected: System notifies the detective that the status has not changed.
Non-Functional Requirements	1. Status transitions must be logged with timestamp and detective identity. 2. Closed cases require Administrator authorization to reopen. 3. Status updates must be reflected in real time for all authorized parties.
Postconditions	1. The case status is updated in the system. 2. The status change is visible in the case timeline and dashboard. 3. Administrator is notified of the status change.

UC15: Write Investigation Report

UC15: Write Investigation Report	
Summary	A Detective writes and submits a formal investigation report summarizing findings, evidence analysis, and investigative conclusions for an assigned case.
Dependency	UC1 (User logs in), UC11 (Open Criminal Case)
Actors	Primary: Detective
Preconditions	1. Detective is logged in. 2. A case is assigned to the detective and has sufficient data for reporting.
Main Sequence	1. Detective navigates to the relevant case. 2. Detective selects 'Write Investigation Report'. 3. System displays the report form. 4. Detective enters: case summary, evidence analysis, suspect assessment, investigative conclusions, and recommended next steps. 5. Detective submits the report. 6. System validates all required fields. 7. System saves the report and links it to the case. 8. System notifies the Administrator of the new report. 9. System confirms with a success message.
Alternative Sequence	1. Missing required fields: System highlights empty fields. 2. Report already exists for the case: System asks the detective to confirm whether to create a new version.
Non-Functional Requirements	1. Reports must be immutable once formally submitted. 2. Reports must be linked to the correct case. 3. All report submissions must be audit-logged with timestamp and detective identity.
Postconditions	1. Investigation report is saved and linked to the case. 2. Report is visible to the Administrator and authorized parties. 3. Submission is recorded in the audit log.

UC16: View and Analyze Case Details

UC16: View and Analyze Case Details	
Summary	A Detective reviews all information associated with a case including incident reports, evidence, witnesses, and suspects.
Dependency	UC1 (User logs in)
Actors	Primary: Detective
Preconditions	1. Detective is logged in. 2. One or more cases are assigned to the detective.
Main Sequence	1. Detective navigates to the Cases section. 2. Detective selects a case assigned to them. 3. System retrieves and displays all case details: crime type, status, assigned officer, suspects, evidence, witnesses, and incident reports. 4. Detective reviews the information. 5. Detective may add notes or initiate follow-up actions (e.g., assign suspect, upload evidence).
Alternative Sequence	1. Case has no data yet: System displays the case with empty sections. 2. Unauthorized access to a case not assigned to the detective: System denies access.
Non-Functional Requirements	1. Case details must load within 3 seconds. 2. All sensitive case data must be accessible only to authorized roles. 3. The view must display all sections clearly: Suspects, Evidence, Witnesses, Incident Reports.
Postconditions	1. Detective has full visibility into all case-related data. 2. Detective can proceed to take further investigative actions from the case view.

4. Forensic Expert

UC17: Perform DNA Analysis

UC17: Perform DNA Analysis	
Summary	A Forensic Expert performs and documents DNA analysis on biological samples submitted as evidence for a case.
Dependency	UC1 (User logs in), UC6 (Upload Evidence)
Actors	Primary: Forensic Expert
Preconditions	1. Forensic Expert is logged in. 2. A DNA sample has been submitted and linked to an active case.
Main Sequence	1. Forensic Expert navigates to the Laboratory section. 2. Expert selects the pending DNA analysis request linked to a case. 3. System displays the analysis form with the sample details. 4. Expert performs the physical DNA analysis in the laboratory. 5. Expert returns to the system and enters the analysis results: methodology, sample condition, findings, and conclusions. 6. Expert submits the analysis. 7. System validates all required fields. 8. System saves the analysis results and links them to the case. 9. System notifies the assigned Detective and Administrator. 10. System confirms with a success message.
Alternative Sequence	1. Sample not yet received: System shows a pending status and prevents analysis entry. 2. Missing required fields: System highlights empty fields.
Non-Functional Requirements	1. DNA analysis records must be immutable once submitted. 2. All analysis data must be stored securely. 3. Results must be audit-logged with timestamp and expert identity.
Postconditions	1. DNA analysis results are saved and linked to the case. 2. Results are visible to authorized parties (Detective, Administrator). 3. Submission is recorded in the audit log.

UC18: Perform Fingerprint Analysis

UC18: Perform Fingerprint Analysis	
Summary	A Forensic Expert analyzes fingerprint samples collected from a crime scene and documents the findings.
Dependency	UC1 (User logs in), UC6 (Upload Evidence)
Actors	Primary: Forensic Expert
Preconditions	1. Forensic Expert is logged in. 2. Fingerprint samples have been submitted and linked to an active case.
Main Sequence	1. Forensic Expert navigates to the Laboratory section. 2. Expert selects the fingerprint analysis request. 3. System displays the fingerprint analysis form with sample details. 4. Expert performs the physical fingerprint analysis. 5. Expert enters findings: print quality, comparison results, identification status, and conclusions. 6. Expert submits the

	analysis. 7. System saves the results and links them to the case. 8. System notifies the assigned Detective and Administrator. 9. System confirms with a success message.
Alternative Sequence	1. Sample too degraded for analysis: Expert enters a 'Non-identifiable' result with notes. 2. Missing required fields: System highlights empty fields.
Non-Functional Requirements	1. Fingerprint analysis records must be immutable once submitted. 2. Comparison results must indicate whether a match was found. 3. All submissions must be audit-logged.
Postconditions	1. Fingerprint analysis results are saved and linked to the case. 2. Results are visible to authorized parties.

UC19: Perform Weapon Analysis

UC19: Perform Weapon Analysis	
Summary	A Forensic Expert performs ballistic, tool mark, and trace evidence analysis on weapons submitted as evidence.
Dependency	UC1 (User logs in), UC6 (Upload Evidence)
Actors	Primary: Forensic Expert
Preconditions	1. Forensic Expert is logged in. 2. A weapon has been submitted and linked to an active case.
Main Sequence	1. Forensic Expert navigates to the Laboratory section. 2. Expert selects the weapon analysis request. 3. System displays the weapon analysis form. 4. Expert performs the physical weapon analysis (ballistics, tool marks, trace evidence). 5. Expert enters findings: weapon type, condition, ballistic results, trace evidence, and conclusions. 6. Expert submits the analysis. 7. System saves the results and links them to the case. 8. System notifies the assigned Detective and Administrator. 9. System confirms with a success message.
Alternative Sequence	1. Weapon not yet received in the lab: System shows pending status. 2. Missing required fields: System highlights empty fields.
Non-Functional Requirements	1. Weapon analysis records must be immutable once submitted. 2. All analysis data must be linked to the correct case. 3. Submissions must be audit-logged.
Postconditions	1. Weapon analysis results are saved and linked to the case. 2. Results are visible to authorized parties.

UC20: Analyze Photos and Videos

UC20: Analyze Photos and Videos	
Summary	A Forensic Expert performs forensic analysis on photos and videos submitted as evidence, including metadata extraction and image enhancement.

Dependency	UC1 (User logs in), UC6 (Upload Evidence)
Actors	Primary: Forensic Expert
Preconditions	1. Forensic Expert is logged in. 2. Photos or videos have been uploaded and linked to an active case.
Main Sequence	1. Forensic Expert navigates to the multimedia evidence section of the relevant case. 2. Expert selects the photo/video analysis request. 3. System displays the analysis form with the uploaded media. 4. Expert performs forensic analysis: metadata extraction, image enhancement, authenticity verification. 5. Expert enters findings and annotates specific frames or regions if applicable. 6. Expert submits the analysis. 7. System saves the results and links them to the case. 8. System notifies the assigned Detective and Administrator. 9. System confirms with a success message.
Alternative Sequence	1. File format not supported for analysis: System notifies the expert and prompts re-upload. 2. File too large for processing: System shows an error and suggests file splitting.
Non-Functional Requirements	1. Analysis results must be stored alongside the original media files. 2. Metadata extraction must be performed automatically where possible. 3. Submissions must be audit-logged with timestamp and expert identity.
Postconditions	1. Photo/video analysis results are saved and linked to the case. 2. Results and annotations are visible to authorized parties.

UC21: Manage Laboratory Evidence

UC21: Manage Laboratory Evidence	
Summary	A Forensic Expert manages the intake, processing, and chain of custody of physical evidence received in the laboratory.
Dependency	UC1 (User logs in)
Actors	Primary: Forensic Expert
Preconditions	1. Forensic Expert is logged in. 2. Physical evidence has been received by the laboratory.
Main Sequence	1. Forensic Expert navigates to the Lab Evidence Management section. 2. Expert clicks 'Register Evidence Intake'. 3. System displays the intake form. 4. Expert enters: evidence ID, case reference, type, condition on arrival, receiving date, and responsible analyst. 5. Expert submits the intake record. 6. System saves the entry and creates a chain of custody record. 7. As the evidence is processed, Expert updates its status (Received, In Analysis, Analysis Complete, Stored). 8. System logs each status change with timestamp and expert identity.
Alternative Sequence	1. Evidence ID already exists: System warns the expert of a potential duplicate. 2. Case reference not found: System shows an error and prevents submission.
Non-Functional Requirements	1. Chain of custody records must be immutable and fully traceable. 2. All status changes must be logged with timestamp and user identity. 3. Evidence records must comply with law enforcement data regulations.

Postconditions	1. Evidence intake is recorded and linked to the case. 2. Chain of custody is established and traceable from receipt to storage. 3. All status transitions are logged in the audit trail.
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UC22: Write Forensic Report

UC22: Write Forensic Report	
Summary	A Forensic Expert writes and submits a formal forensic report summarizing all analysis results for a specific case.
Dependency	UC1 (User logs in), UC17/UC18/UC19/UC20 (Any analysis)
Actors	Primary: Forensic Expert
Preconditions	1. Forensic Expert is logged in. 2. At least one analysis has been completed and linked to the case.
Main Sequence	1. Forensic Expert navigates to the relevant case. 2. Expert selects 'Write Forensic Report'. 3. System displays the report form, pre-populated with linked analyses. 4. Expert enters: report summary, methodology description, all analysis findings, conclusions, and case implications. 5. Expert submits the report. 6. System validates all required fields. 7. System saves the report and links it to the case. 8. System notifies the assigned Detective and Administrator. 9. System confirms with a success message.
Alternative Sequence	1. No completed analyses linked to the case: System warns the expert that at least one analysis must be complete. 2. Missing required fields: System highlights empty fields.
Non-Functional Requirements	1. Forensic reports must be immutable once formally submitted. 2. Reports must be linked to all related analyses. 3. Submissions must be audit-logged.
Postconditions	1. Forensic report is saved and linked to the case. 2. Report is visible to the Detective and Administrator. 3. Submission is recorded in the audit log.

5. Traffic Officer

UC23: Register Traffic Accident

UC23: Register Traffic Accident	
Summary	A Traffic Officer registers a traffic accident, documenting vehicles involved, injuries, witnesses, and location.
Dependency	UC1 (User logs in)
Actors	Primary: Traffic Officer
Preconditions	1. Traffic Officer is logged in. 2. A traffic accident has occurred and been reported.
Main Sequence	1. Officer navigates to the Traffic Accidents section. 2. Officer clicks 'Register Accident'. 3. System displays the accident registration form. 4. Officer enters: date, time, location, vehicle details (registration, make, model), number of people involved, injury description, witness information, and accident description. 5. Officer submits the form. 6. System validates all required fields. 7. System saves the accident record. 8. System generates an accident report ID. 9. System confirms with a success message.
Alternative Sequence	1. Missing required fields: System highlights empty fields. 2. System save error: System displays an error and prompts retry.
Non-Functional Requirements	1. Accident records must be immutable once submitted (controlled edit process). 2. Data must comply with traffic law and data protection regulations. 3. Records must be audit-logged with timestamp and officer identity.
Postconditions	1. Accident record is saved and assigned a unique ID. 2. Accident is visible in the Traffic Accidents list. 3. Record is accessible for reporting and statistics.

UC24: Issue Traffic Fine

UC24: Issue Traffic Fine	
Summary	A Traffic Officer issues and registers a fine for a traffic violation to a citizen or vehicle.
Dependency	UC1 (User logs in)
Actors	Primary: Traffic Officer
Preconditions	1. Traffic Officer is logged in. 2. A traffic violation has been observed or reported.
Main Sequence	1. Officer navigates to the Fines section. 2. Officer clicks 'Issue Fine'. 3. System displays the fine registration form. 4. Officer enters: violation type, fine amount, date, time, location, citizen ID or vehicle registration, and description. 5. Officer submits the form. 6. System validates all required fields. 7. System saves the fine record and links it to the citizen

	or vehicle. 8. System makes the fine visible to the citizen in their portal. 9. System confirms with a success message.
Alternative Sequence	1. Citizen or vehicle not found in the system: System allows the officer to manually enter the details. 2. Missing required fields: System highlights empty fields.
Non-Functional Requirements	1. Fine records must be immutable once issued. 2. Fines must be immediately visible to the citizen in their portal. 3. All fine issuances must be audit-logged.
Postconditions	1. Fine is saved and linked to the citizen or vehicle. 2. Fine is visible to the citizen in their portal. 3. Fine issuance is recorded in the audit log.

UC25: Manage Road Violations

UC25: Manage Road Violations	
Summary	A Traffic Officer records, views, and manages road violations and infractions.
Dependency	UC1 (User logs in)
Actors	Primary: Traffic Officer
Preconditions	1. Traffic Officer is logged in. 2. A road violation has been observed or reported.
Main Sequence	1. Officer navigates to the Violations section. 2. To register a violation: Officer clicks 'Register Violation', enters violation type, location, date, vehicle details, and description. Officer submits the form. System saves the violation record. 3. To view violations: Officer browses or filters the violations list by date, type, or status. 4. To update a violation: Officer selects a violation, modifies the status (Open, Resolved), and saves. 5. System confirms each action with a success message.
Alternative Sequence	1. Missing required fields: System highlights empty fields. 2. Violation already closed: System prevents re-opening without authorization.
Non-Functional Requirements	1. Violation records must be saved within 2 seconds. 2. Closed violations must be audit-logged before status change. 3. Access control prevents unauthorized modifications.
Postconditions	1. Violation record is saved and accessible in the Violations list. 2. Status updates are reflected immediately. 3. All changes are audit-logged.

UC26: Perform Vehicle Control

UC26: Perform Vehicle Control	
Summary	A Traffic Officer performs and documents a vehicle control check, verifying registration, insurance, and inspection status.
Dependency	UC1 (User logs in)
Actors	Primary: Traffic Officer

Preconditions	1. Traffic Officer is logged in. 2. A vehicle has been stopped for a control check.
Main Sequence	1. Officer navigates to the Vehicle Control section. 2. Officer clicks 'New Vehicle Control'. 3. System displays the control form. 4. Officer enters: vehicle registration number, make, model, year, owner details, date and time of check. 5. Officer checks registration status, insurance validity, and inspection certificate. 6. Officer enters the control results and any notes. 7. Officer submits the form. 8. System saves the control record and links it to the vehicle. 9. System confirms with a success message.
Alternative Sequence	1. Vehicle registration not found: System allows manual entry of vehicle details. 2. Missing required fields: System highlights empty fields.
Non-Functional Requirements	1. Vehicle control records must be immutable once submitted. 2. Results must be clearly linked to the vehicle and date of control. 3. All control records must be audit-logged.
Postconditions	1. Vehicle control record is saved and linked to the vehicle. 2. Record is accessible in the Vehicle Controls list. 3. Control results are audit-logged.

UC27: Manage Driving Licenses

UC27: Manage Driving Licenses	
Summary	A Traffic Officer searches, views, and manages driving license records, including flagging suspended or expired licenses.
Dependency	UC1 (User logs in)
Actors	Primary: Traffic Officer
Preconditions	1. Traffic Officer is logged in.
Main Sequence	1. Officer navigates to the Licenses section. 2. Officer searches for a license by citizen ID, license number, or name. 3. System retrieves and displays the license record: license number, category, issue date, expiry date, status. 4. Officer reviews the license validity. 5. To flag a license: Officer selects 'Flag / Suspend', enters the reason, and confirms. System updates the license status. 6. System confirms each action with a success message.
Alternative Sequence	1. License not found: System displays a 'No license record found' message. 2. License already suspended: System notifies the officer.
Non-Functional Requirements	1. License status changes must be audit-logged with timestamp and officer identity. 2. Citizens must be notified if their license is flagged or suspended. 3. License data must be stored securely with role-restricted access.
Postconditions	1. License status is updated and visible in the system. 2. Citizen is notified of any flag or suspension. 3. Status change is audit-logged.

UC28: Generate Traffic Report

UC28: Generate Traffic Report	
Summary	A Traffic Officer generates statistical reports covering violations, accidents, and fine collection for a specified period.
Dependency	UC1 (User logs in)
Actors	Primary: Traffic Officer Secondary: Administrator
Preconditions	1. Traffic Officer is logged in. 2. Traffic data (accidents, violations, fines) exists in the system.
Main Sequence	1. Officer navigates to the Traffic Reports section. 2. Officer selects the report type: Violations Summary, Accident Statistics, or Fine Collection Report. 3. Officer specifies the date range and any additional filters. 4. Officer clicks 'Generate Report'. 5. System processes the data. 6. System displays the report with summary statistics and optional visualizations. 7. Officer can export the report (PDF or CSV).
Alternative Sequence	1. No data available for the selected period: System displays a 'No data found' message. 2. Export failure: System shows an error and allows retry.
Non-Functional Requirements	1. Reports must generate within 10 seconds. 2. Reports must be accurate and sourced from live database data. 3. Exported files must be formatted clearly and comply with department standards.
Postconditions	1. Traffic report is generated and displayed. 2. Report can be exported if needed. 3. Report generation is logged in the audit trail.

6. Citizen

UC29: Self-Register as Citizen

UC29: Self-Register as Citizen	
Summary	A member of the public creates a citizen account on the PMS portal to access public-facing services.
Dependency	None
Actors	Primary: Citizen
Preconditions	1. The citizen portal is publicly accessible. 2. The person does not already have a citizen account.
Main Sequence	1. Citizen navigates to the public PMS portal. 2. Citizen clicks 'Register'. 3. System displays the citizen registration form. 4. Citizen enters: full name, national ID number, date of birth, email address, phone number, and password. 5. Citizen submits the form. 6. System validates all required fields and checks for duplicate accounts. 7. System creates the citizen account. 8. System sends a confirmation notification or email. 9. System confirms with a success message.
Alternative Sequence	1. Duplicate account (same national ID or email): System notifies the citizen and prevents duplicate registration. 2. Missing required fields: System highlights empty fields. 3. Weak password: System requires a stronger password.
Non-Functional Requirements	1. Registration must complete within 5 seconds. 2. Citizen account data must be stored securely and comply with GDPR. 3. Citizens must only have access to their own submitted data.
Postconditions	1. Citizen account is created and active. 2. Citizen can log in and access the citizen portal. 3. Registration is recorded in the system.

UC30: Report an Incident

UC30: Report an Incident	
Summary	A Citizen submits an incident report through the public portal to notify the police of a crime or suspicious activity.
Dependency	UC1 (Citizen logs in)
Actors	Primary: Citizen
Preconditions	1. Citizen is logged in to their portal account. 2. The incident has occurred.
Main Sequence	1. Citizen navigates to the 'Report Incident' section. 2. Citizen clicks 'Submit Report'. 3. System displays the incident report form. 4. Citizen enters: incident description, location, date and time, and optionally uploads a photo. 5. Citizen submits the form. 6. System validates all required fields. 7. System saves the incident report. 8. System assigns the report to the police department for review. 9. System confirms with a success message and provides a reference number.

Alternative Sequence	1. Missing required fields: System highlights empty fields. 2. File upload failure: System retains the form data and prompts retry for the file only.
Non-Functional Requirements	1. Incident reports must be saved within 3 seconds. 2. Citizens must receive a reference number for tracking. 3. Report data must be stored securely.
Postconditions	1. Incident report is saved and submitted to the police department. 2. Citizen receives a reference number for tracking. 3. Report is accessible in the citizen's portal for status tracking.

UC31: Submit a Complaint

UC31: Submit a Complaint	
Summary	A Citizen submits a formal complaint regarding police conduct, services, or a specific officer through the public portal.
Dependency	UC1 (Citizen logs in)
Actors	Primary: Citizen
Preconditions	1. Citizen is logged in to their portal account. 2. The citizen has a complaint to submit.
Main Sequence	1. Citizen navigates to the 'Submit Complaint' section. 2. Citizen clicks 'New Complaint'. 3. System displays the complaint form. 4. Citizen enters: complaint category, description, date of the incident, and optionally the relevant officer or department. 5. Citizen submits the form. 6. System validates all required fields. 7. System saves the complaint and routes it to the Administrator. 8. System confirms with a success message and provides a complaint reference number.
Alternative Sequence	1. Missing required fields: System highlights empty fields. 2. System save error: System retains the form and prompts retry.
Non-Functional Requirements	1. Complaints must be saved within 3 seconds. 2. Citizens must receive a reference number. 3. All complaints must be routed to the Administrator for review.
Postconditions	1. Complaint is saved and routed to the Administrator. 2. Citizen receives a reference number. 3. Complaint is accessible in the citizen's portal for status tracking.

UC32: Report a Missing Person

UC32: Report a Missing Person	
Summary	A Citizen submits a missing person report through the public portal.
Dependency	UC1 (Citizen logs in)
Actors	Primary: Citizen
Preconditions	1. Citizen is logged in to their portal account. 2. A person has gone missing.

Main Sequence	1. Citizen navigates to the 'Report Missing Person' section. 2. Citizen clicks 'File Report'. 3. System displays the missing person report form. 4. Citizen enters: missing person's name, date of birth, physical description, last known location, date last seen, and citizen's contact information. 5. Citizen optionally uploads a recent photo. 6. Citizen submits the form. 7. System validates all required fields. 8. System saves the report and assigns it to the relevant police department. 9. System confirms with a success message and provides a reference number.
Alternative Sequence	1. Missing required fields: System highlights empty fields. 2. Photo upload failure: System retains the form and prompts retry.
Non-Functional Requirements	1. Missing person reports must be saved within 3 seconds. 2. Citizens must receive a reference number. 3. Reports must be immediately accessible to the assigned department.
Postconditions	1. Missing person report is saved and submitted to the police department. 2. Citizen receives a reference number. 3. Report is accessible in the citizen's portal for status tracking.

UC33: Check Case Status

UC33: Check Case Status	
Summary	A Citizen checks the current status of an incident report, complaint, or missing person report they have submitted.
Dependency	UC1 (Citizen logs in), UC30/UC31/UC32
Actors	Primary: Citizen
Preconditions	1. Citizen is logged in. 2. The citizen has at least one submitted report in the system.
Main Sequence	1. Citizen navigates to 'My Reports' section. 2. System retrieves all reports submitted by the citizen. 3. System displays the list with: report type, reference number, date submitted, and current status. 4. Citizen selects a specific report. 5. System displays the full report details and current status. 6. Citizen reviews the status update.
Alternative Sequence	1. No reports found: System displays a message 'You have not submitted any reports'. 2. Report status not yet updated: System displays the last known status with a 'Pending Review' label.
Non-Functional Requirements	1. Report list must load within 2 seconds. 2. Citizens can only view their own submitted reports. 3. Status information must reflect the most recent update.
Postconditions	1. Citizen can view the current status of all their submitted reports. 2. Citizen is informed of any updates or next steps.

UC34: View Traffic Fines

UC34: View Traffic Fines

Summary	A Citizen views any traffic fines issued to them by a Traffic Officer through the citizen portal.
Dependency	UC1 (Citizen logs in)
Actors	Primary: Citizen
Preconditions	1. Citizen is logged in. 2. At least one fine has been issued to the citizen by a Traffic Officer.
Main Sequence	1. Citizen navigates to the 'My Fines' section. 2. System retrieves all fines linked to the citizen's account. 3. System displays the list with: fine reference, violation type, amount, date issued, and payment status. 4. Citizen selects a specific fine. 5. System displays the full fine details: issuing officer, location, violation description, and deadline.
Alternative Sequence	1. No fines found: System displays a message 'No fines have been issued to your account'. 2. Fine data loading error: System shows an error and provides a refresh option.
Non-Functional Requirements	1. Fine list must load within 2 seconds. 2. Citizens can only view their own fines. 3. Fine data must be accurate and reflect the most recent status.
Postconditions	1. Citizen can view all traffic fines issued to them. 2. Citizen is informed of the violation type, amount, and payment status.

7. Administrator

UC35: Manage Users and Roles

UC35: Manage Users and Roles	
Summary	The Administrator creates, updates, deactivates, and manages user accounts and role assignments for all staff roles (Police Officer, Detective, Forensic Expert, Traffic Officer).
Dependency	UC1 (Administrator logs in)
Actors	Primary: Administrator
Preconditions	1. Administrator is logged in. 2. The system is operational.
Main Sequence	1. Administrator navigates to the User Management section. 2. To add a user: Administrator clicks 'Add User', fills in Full Name, Username, Password, Role (Officer/Detective/Forensic Expert/Traffic Officer), and Department. Clicks Save. System validates and creates the account. 3. To edit a user: Administrator selects a user, clicks 'Edit', modifies the required fields, and saves. 4. To deactivate a user: Administrator selects a user and clicks 'Deactivate'. System prompts for confirmation. Administrator confirms. System deactivates the account. 5. System confirms each action with a success message.
Alternative Sequence	1. Duplicate username: System prevents creation and displays an error. 2. Missing required fields: System highlights missing fields. 3. Attempt to deactivate an active user with ongoing cases: System warns the administrator and requests confirmation.

Non-Functional Requirements	1. User management operations must complete within 3 seconds. 2. Passwords must be hashed before storage. 3. All user management actions must be audit-logged.
Postconditions	1. New user account is created with the correct role and department. 2. Updated user information is reflected in the system. 3. Deactivated user can no longer log in.

UC36: Manage Departments

UC36: Manage Departments	
Summary	The Administrator adds, edits, and manages police department records to organize staff by their respective units.
Dependency	UC1 (Administrator logs in)
Actors	Primary: Administrator
Preconditions	1. Administrator is logged in. 2. System is operational.
Main Sequence	1. Administrator navigates to the Departments section. 2. To add: Administrator clicks 'Add', enters the Department Name, and saves. 3. To edit: Administrator selects a department, clicks 'Edit', modifies the name, and saves. 4. To delete: Administrator selects a department and clicks 'Delete'. System checks if staff are assigned. If no staff assigned, department is deleted. If staff are assigned, system warns the administrator. 5. System confirms each action with a success message.
Alternative Sequence	1. Department name already exists: System displays a duplicate error. 2. Attempt to delete department with assigned staff: System warns and prevents deletion until staff are reassigned.
Non-Functional Requirements	1. Department operations should complete within 2 seconds. 2. Department names must be unique within the system. 3. Changes must be immediately visible to all relevant users.
Postconditions	1. New department is saved and available for staff assignment. 2. Updated department information is reflected in the system. 3. Deleted department is removed from all dropdown lists.

UC37: Assign Officer to Case

UC37: Assign Officer to Case	
Summary	The Administrator assigns a Police Officer to a specific case to ensure proper investigative coverage.
Dependency	UC1 (Administrator logs in), UC4 (Create Incident Report)
Actors	Primary: Administrator
Preconditions	1. Administrator is logged in. 2. An active case exists in the system. 3. One or more Police Officers are registered in the system.
Main Sequence	1. Administrator navigates to the active case. 2. Administrator selects 'Assign Officer'. 3. System displays a list of available Police Officers. 4.

	Administrator selects the appropriate officer. 5. Administrator confirms the assignment. 6. System validates the selection. 7. System saves the assignment and notifies the assigned officer. 8. System confirms with a success message.
Alternative Sequence	1. No available officers: System displays a message indicating all officers are currently assigned. 2. Case already has an assigned officer: System asks the Administrator to confirm the reassignment.
Non-Functional Requirements	1. Assignment must be saved and reflected within 2 seconds. 2. The assigned officer must receive a notification. 3. Officer assignment history must be logged.
Postconditions	1. The selected officer is assigned to the case. 2. The case details are updated with the assigned officer's information. 3. The officer can now access and manage the case.

UC38: Monitor System Statistics

UC38: Monitor System Statistics	
Summary	The Administrator monitors real-time analytics and statistics across the entire system including cases, users, departments, forensic activity, and traffic data.
Dependency	UC1 (Administrator logs in), UC2 (View Dashboard)
Actors	Primary: Administrator
Preconditions	1. Administrator is logged in. 2. Data has been entered into the system by users across all roles.
Main Sequence	1. Administrator navigates to the Dashboard. 2. System retrieves all system-wide statistics. 3. System displays: number of active departments, users per role, open cases, cases under investigation, recent incidents, pending forensic analyses, active traffic violations, and recent fines. 4. Administrator reviews the data. 5. Administrator can navigate to specific sections for more detailed information.
Alternative Sequence	1. No data available: System displays zero values and a note indicating the system has no recorded data. 2. Data loading failure: System shows an error and provides a refresh option.
Non-Functional Requirements	1. Dashboard statistics must reflect real-time data updated within 1 minute. 2. Statistics page must load within 5 seconds. 3. Data displayed must be accurate and sourced from the live database.
Postconditions	1. Administrator has a comprehensive, real-time overview of all system operations. 2. Administrator can take corrective action based on the statistics displayed.

UC39: Reset User Password

UC39: Reset User Password

Summary	The Administrator resets the password of a staff user account when requested or necessary.
Dependency	UC1 (Administrator logs in), UC35 (Manage Users and Roles)
Actors	Primary: Administrator
Preconditions	1. Administrator is logged in. 2. The target user account exists in the system.
Main Sequence	1. Administrator navigates to the Users section. 2. Administrator selects the target user. 3. Administrator clicks 'Reset Password'. 4. System prompts for confirmation. 5. Administrator confirms. 6. System generates a temporary password and updates the user's account. 7. System notifies the user via the system or email with the temporary credentials. 8. System confirms with a success message.
Alternative Sequence	1. User account not found: System displays an error. 2. Notification delivery failure: System logs the failure and alerts the administrator.
Non-Functional Requirements	1. Temporary password must meet minimum security requirements. 2. Password reset must be logged in the audit trail. 3. User must be required to change the temporary password on next login.
Postconditions	1. User's password is reset to a temporary value. 2. User is notified of the reset. 3. The reset action is recorded in the audit log.

UC40: View System Audit Log

UC40: View System Audit Log	
Summary	The Administrator views a log of all significant actions performed in the system for accountability and security monitoring.
Dependency	UC1 (Administrator logs in)
Actors	Primary: Administrator
Preconditions	1. Administrator is logged in. 2. The system has recorded at least one auditable action.
Main Sequence	1. Administrator navigates to the Audit Log section. 2. System retrieves all logged actions from the database. 3. System displays the log: action type, user role, username, timestamp, and affected record. 4. Administrator can filter logs by user, date range, role, or action type. 5. Administrator reviews the audit entries.
Alternative Sequence	1. No logs available: System displays a message 'No audit records found'. 2. Filter returns no results: System displays 'No results match the selected filters'.
Non-Functional Requirements	1. Audit log must be read-only --- no entries can be edited or deleted. 2. Log entries must include user identity, timestamp, and action description. 3. Audit log must load within 3 seconds.
Postconditions	1. Administrator has visibility into all audited system actions. 2. Administrator can identify any unauthorized or suspicious activity.

UC41: Search Officers by Department

UC41: Search Officers by Department	
Summary	The Administrator searches for and filters staff members based on their department assignment or role.
Dependency	UC1 (Administrator logs in), UC36 (Manage Departments)
Actors	Primary: Administrator
Preconditions	1. Administrator is logged in. 2. At least one department and one staff member exist in the system.
Main Sequence	1. Administrator navigates to the Users section. 2. Administrator selects a department from the filter dropdown or enters a role filter. 3. System queries the database for staff in the selected department or role. 4. System displays the filtered list of staff members with their ID, name, role, and department. 5. Administrator can select a staff member to view or edit their profile.
Alternative Sequence	1. No staff in selected department: System displays 'No staff found in this department'. 2. Invalid filter selection: System resets to show all staff.
Non-Functional Requirements	1. Search results must be returned within 2 seconds. 2. Results must reflect the current state of the database. 3. Only authorized administrators can access staff data.
Postconditions	1. A filtered list of staff matching the selected criteria is displayed. 2. Administrator can take action on any staff member from the results.

UC42: Deactivate User Account

UC42: Deactivate User Account	
Summary	The Administrator deactivates a staff user account to revoke system access without permanently deleting the account and its data.
Dependency	UC1 (Administrator logs in), UC35 (Manage Users and Roles)
Actors	Primary: Administrator
Preconditions	1. Administrator is logged in. 2. The target user account exists and is currently active.
Main Sequence	1. Administrator navigates to the Users section. 2. Administrator selects the target user. 3. Administrator clicks 'Deactivate Account'. 4. System prompts for confirmation and optionally a reason. 5. Administrator enters the reason and confirms. 6. System sets the account status to Inactive. 7. System terminates any active session for the deactivated user. 8. System logs the deactivation with the reason and timestamp. 9. System confirms with a success message.
Alternative Sequence	1. User has ongoing open cases: System warns the administrator and requests confirmation before proceeding. 2. Administrator cancels: Account remains active.

Non-Functional Requirements	1. Deactivation must take effect immediately and terminate the user's active session. 2. Deactivation reason must be stored in the audit log. 3. Deactivated account data must be preserved for historical integrity.
Postconditions	1. The user account is set to Inactive and access is revoked. 2. The user can no longer log in. 3. All account data is preserved and remains accessible to administrators. 4. Deactivation is recorded in the audit log with reason and timestamp.